69.

Solution:

Given :

$T = 298\ K$

$\Delta H = 400\ KJ\ mol^{-1}$

$\Delta S = 0.2\ KJ\ K^{-1}\ mol^{-1}$

$\Delta G = \Delta H - T\Delta S$

if $T = 2000\ K$

$\Delta G = 400 - (0.2 \times 2000) = 0$

$\Delta H = 400\ KJ\ mol^{-1}$ if $T > 2000\ K$

ΔG will be negative.

The reaction would be spontaneous only beyond 2000K

70.

Solution :

Given :

$T = 298K$

$\Delta G_r^0 = -13.6\ kJ\ mol^{-1}$

$= -13600\ J\ mol^{-1}$

$\Delta G^0 = -2.303\ RT \log K_{eq}$

$$\log K_{eq} = \frac{-\Delta G^0}{2.303\ RT} \quad \log K_{eq} = \frac{13.6 kJ\ mol^{-1}}{2.303 \times 8.314 \times 10^{-3} JK^{-1}\ mol^{-1} \times 298\ K}$$

$\log K_{eq} = 2.38$

$K_{eq} = \text{antilog } (2.38)$

$K_{eq} = 239.88$

71.

Solution :

Given :

$\Delta H_C\ (CH_4) = -890\ kJ\ mol^{-1}$

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